Turfify
PLAY. BOOK. ENJOY.

PRODUCTION TECHNICAL SPECIFICATION

COMPREHENSIVE ENGINEERING GUIDE & CLIENT BLUEPRINT

Frontend Architecture & Interactive UI/UX Specification

An exhaustive, component-by-component manual detailing every frontend screen, navigation button, search parameter, interactive click reaction, modal popup, and dual-role operational workflow.

PLATFORM VERSION	TARGET ECOSYSTEM	ARCHITECTURE MODEL	STATE MANAGEMENT
v1.0 Production	Web & Mobile Responsive	Player Portal + Owner ERP	TurfifyStore (Supabase Ready)

1. Consumer Marketplace
Real-time location, sport, date, and hourly slot filtering with instant booking checkout modal.

2. Player Hub Portal
Live kickoff countdown, GPS Google Maps navigation, WhatsApp squad share, and match ledger.

3. Turf Owner ERP
Real-time daily turnover metrics, slot occupancy matrix, walk-in offline blocking, and financial settlements.

Document Prepared For: **Executive Client Review & Engineering Alignment**

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1. Platform Architecture & Master Page Inventory

The Turfify frontend is architected as a **high-performance, dual-role sports ecosystem**. It provides both consumer-facing match booking for active sports enthusiasts and an enterprise facility console for turf owners.

> **Core Architectural Principle: Zero Double-Booking & Instant UI Responsiveness**
> Every interactive button communicates through the unified `TurfifyStore` state machine. When a player books an hourly slot or an owner logs a cash walk-in, the slot is locked immediately across all views to prevent slot collisions.

1.1 Master Page Inventory

Page / Screen	File Path	Primary User Role	Key Functional Purpose
Marketplace Home	`index.html`	Public / Player	Hero banner, multi-parameter search bar, sport pills, verified turf listings, owner registration banner.
Player Hub Portal	`player-dashboard.html`	Registered Player	Next match kickoff countdown, GPS directions, squad sharing, active bookings, match cancellation, loyalty coins.
Turf Owner ERP	`owner-dashboard.html`	Turf Facility Owner	Today's turnover, slot occupancy rate, 16-hour live slot matrix, manual walk-in blocker, add grounds modal, payouts.
Unified Portal	`dashboard.html`	Dual-Role Switcher	Seamless toggle between Player View and Owner View without page reloading or loss of state.
About Platform	`about.html`	Public / Enterprise	Company background, sports infrastructure mission, founder vision, investor metrics, and partner contact.

1.2 Modal Overlay Architecture

The frontend uses three non-destructive modal overlays that mount dynamically over any viewport:

Slot Booking Modal PLAYER

Pre-fills venue pricing, calendar date picker, morning/evening slot chips, dynamic price calculation, and instant reservation trigger.

Owner Facility Modal PARTNER

Pitches onboarding form: venue name, sport type, ground dimensions, price per hour, and phone number for 2-hour onboarding dispatch.

Authentication Modal DUAL ROLE

Interactive role selection toggle (Player vs. Turf Owner) allowing dynamic redirect to either Player Hub or Partner Console.

2. Consumer Marketplace: Landing & Hero Viewport

The Marketplace Home Page (`index.html`) provides high-conversion discovery with trust badges, quick booking access, and dedicated owner acquisition.

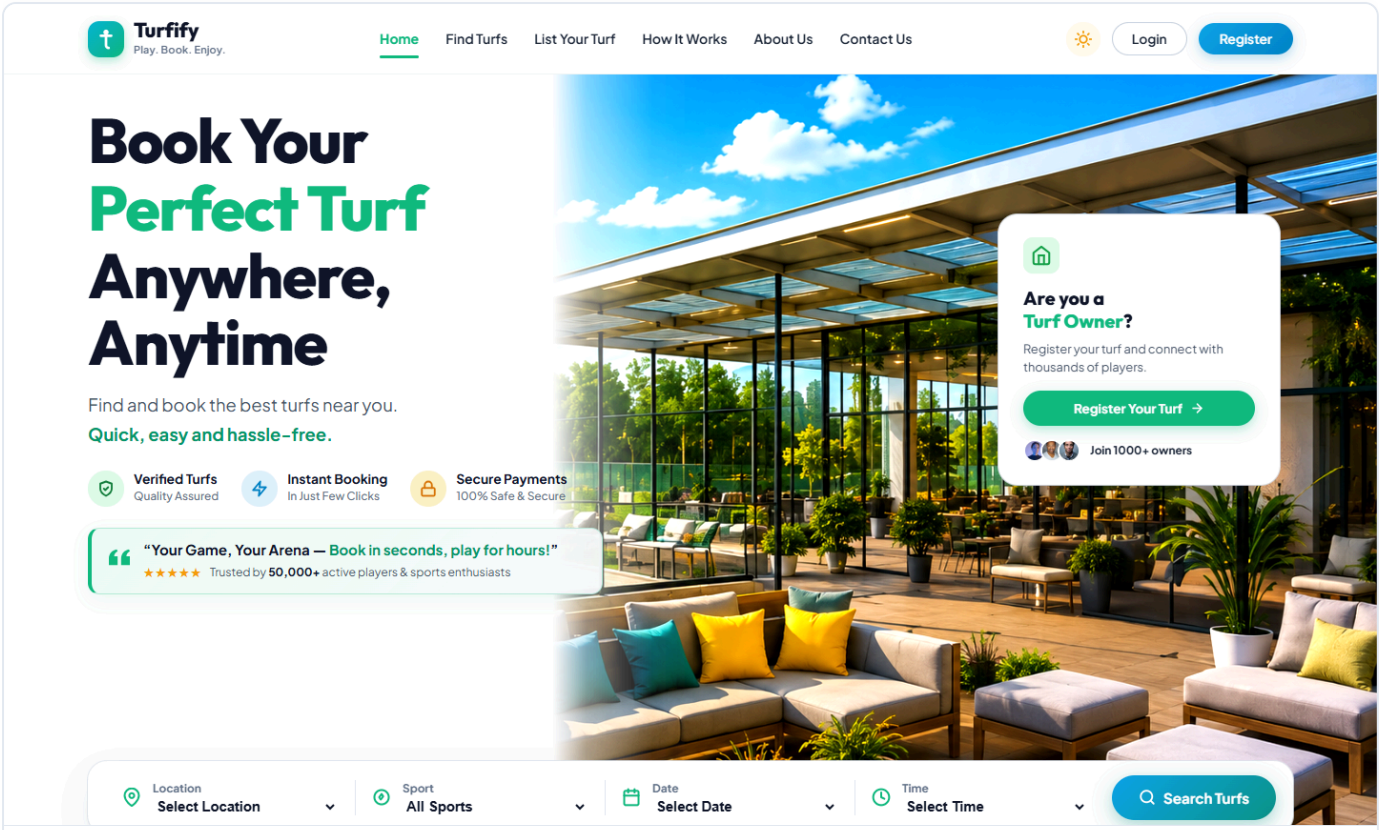


Figure 2.1 — Marketplace Home Viewport with Navigation, Trust Badges, and Search Console PRODUCTION UI

2.1 Navigation Bar & Header Actions

Navbar Element	Trigger Type	What Happens on Click
Brand Logo ("Turfify")	Link (`href="#home"`)	Smoothly scrolls viewport back to the top hero section.
"Find Turfs" Link	Link (`href="#search-section"`)	Smooth-scrolls viewport to the search console and focuses search fields.
"Dashboard" Link	Nav Action (`dashboard.html`)	Navigates directly to the interactive portal based on the active logged-in profile.
"List Your Turf" Link	Modal Trigger (`#btn-hero-owner`)	Opens the **Turf Owner Registration Modal** for ground partners.
Theme Toggle Button	Button (`#btn-theme-toggle`)	Toggles high-contrast daylight mode and displays toast notification.
"Login" & "Register"	Auth Trigger (`#btn-open-login`)	Pops open the **Authentication Modal** with Player / Owner role toggles.

3. The Interactive Search Engine: Exact "Click → Reaction" System

Detailed engineering breakdown of user clicks across **Location, Sport, Date, and Time**:



Figure 3.1 — Multi-Parameter Search Console (Location, Sport, Date, Time & Search CTA) INTERACTIVE COMPONENT

3.1 Click-by-Click Parameter Behavior Matrix

Filter Parameter	User Action (What They Click)	What the Frontend Shows (Visual UI)	Technical State & System Reaction
1. Location Dropdown `#select-location`	Selects area from dropdown: • Koramangala, HSR Layout, Indiranagar, Jayanagar, Whitefield, All.	• Dropdown displays selected neighborhood. • Mismatched cards fade out instantly. • Animated toast: *"Showing X verified venues matching your selection."*	• Evaluates `card.dataset.location`. • Fallback when 0 match: Displays all premier venues with notification toast.
2. Sport Selector `#select-sport`	Selects sport: • Football, Cricket, Badminton, Basketball.	• Active sport pill card highlights in pastel tint. • Rerenders venue list with sport amenities. • Smoothly scrolls user to `#popular-turfs`.	• Matches `card.dataset.sport === sport`. • Syncs selection across search bar and sports pills.
3. Date Selector `#select-date`	Selects date: • Today, Tomorrow, Weekend, Calendar.	• Updates search bar date label. • Adjusts slot counters on cards (*"12 slots left today"*). • Passes date into the Booking Modal.	• Queries `TurfifyStore.getSlots(turfId, date)`. • Excludes slots booked by players or blocked by owner.
4. Time Window `#select-time`	Selects hourly quadrant: • Morning (6–10 AM), Day (10–4 PM), Evening (4–8 PM), Night (8–Midnight).	• Shows venues operating during that window. • Evening/Night prioritizes **Floodlights**. • Day prioritizes Covered / AC venues.	• Slot-level filter against facility operating hours. • Sorts search results by highest availability.
5. Search Turfs Button `#btn-search-turfs`	Clicks teal gradient search pill button.	• Button exhibits tactile press effect. • Viewport smooth-scrolls to venue listings. • Notification toast confirms matching venues.	• Intercepts form submit via `e.preventDefault()`. • Compiles composite filter: `Location + Sport + Date + Time`.

4. Sports Category Pills & Interactive Booking Modal

In addition to the search bar, users can click any of the **5 Sport Category Cards** for one-click discovery:



Figure 4.1 — 5 Sport Category Filter Pills (Football, Cricket, Badminton, Basketball, Swimming) ONE-CLICK FILTER

4.1 Venue Card Anatomy

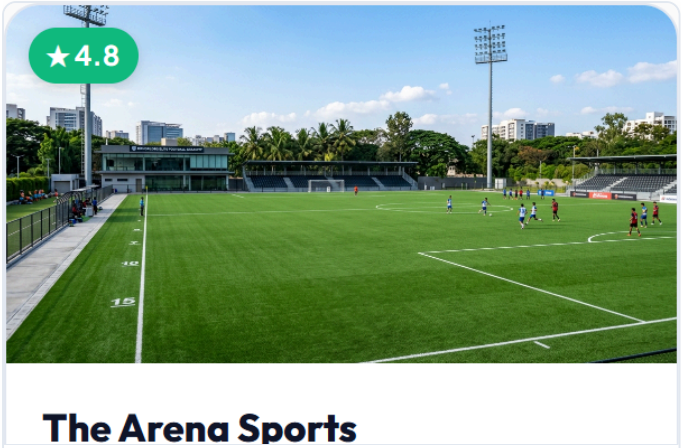


Figure 4.2 — Verified Turf Card

- **Favorite Heart**: Toggles red heart & adds to favorites.
- **Book Now**: Opens live booking modal with pre-filled rates.

4.2 Live Booking Modal

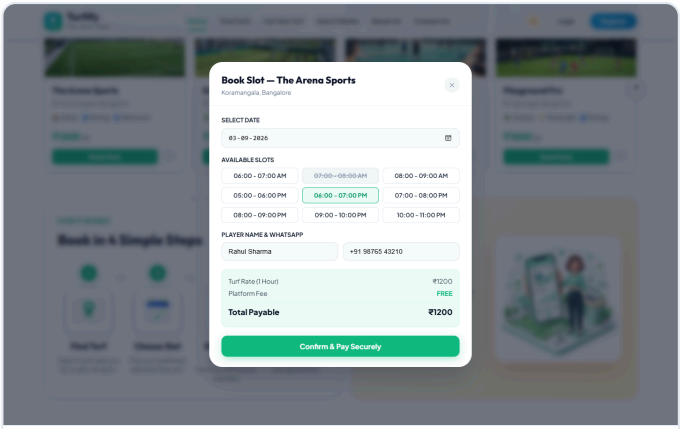
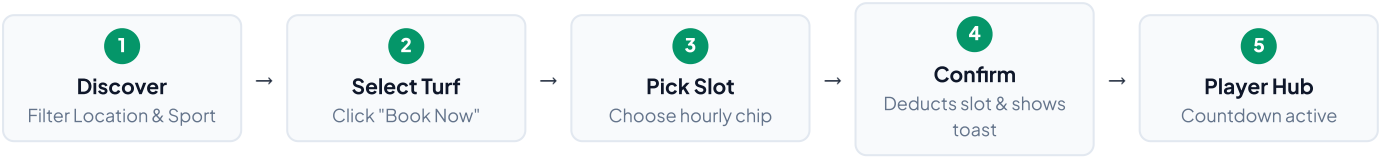


Figure 4.3 — Slot Booking Modal

- **Open Slot Chip**: Turns emerald green; updates price.
- **Confirm & Pay**: Writes to store; notifies user via toast.

4.3 Complete Booking Sequence



5. Player Hub: Match Management, GPS & Squad Sharing

The Player Hub (`player-dashboard.html`) is the player's personal command center for tracking matches, sharing invite links with teammates, and managing cancellations.

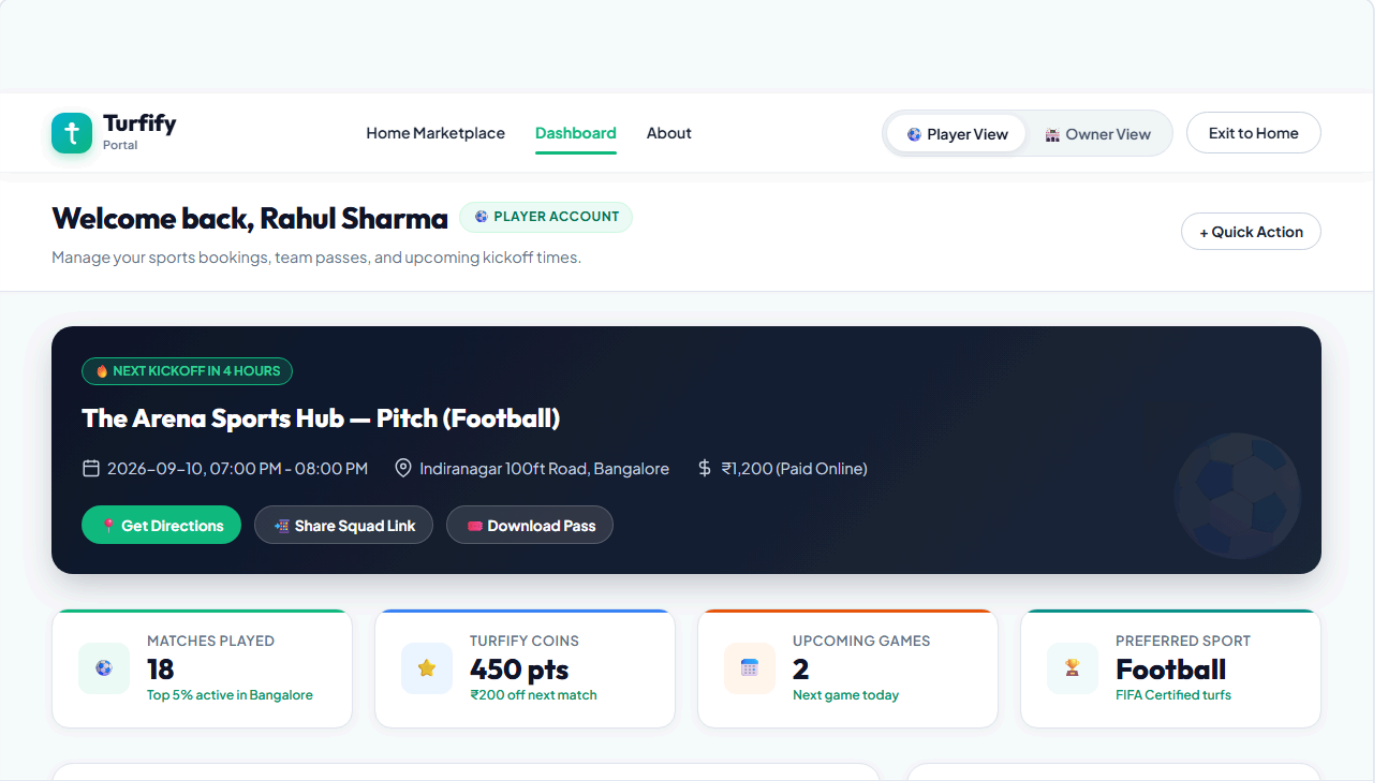


Figure 5.1 — Player Hub with Kickoff Countdown Card, Active Match Ledger, and Navigation Actions PLAYER CONSOLE

5.1 Player Interactive Feature Breakdown

Component	Button / Element	User Action & Click Reaction
Next Kickoff Hero Card	📍 **Get Directions** `btn-match-primary`	Launches **Google Maps GPS navigation** with exact ground coordinates (e.g. Indiranagar 100ft Road Arena), providing driving time and turn-by-turn routing.
Squad Coordination	📲 **Share Squad Link** `btn-match-secondary`	Copies encrypted WhatsApp squad invite link to clipboard. Teammates clicking the link can see match time, pitch number, kit color, and ground rules.
Match Modification	❌ **Cancel Match** `btn-cancel-match`	Prompts user with refund policy modal. Upon confirmation, marks booking status as `Cancelled` in `TurfifyStore` and releases the slot back to the public marketplace.
Bookings Ledger	**My Matches Table**	Displays full history of all past and upcoming matches with status tags: `Paid Online`, `Confirmed`, `Completed`, or `Walk-in`.
Gamification & Loyalty	**Loyalty Coins Badge**	Displays active balance (e.g. **450 Coins**). Every confirmed match grants 25 coins, redeemable for discounts during peak evening slots.

6. Turf Owner Partner Console: Business Metrics & Slot Matrix

The Partner Console (`owner-dashboard.html`) provides turf facility owners with live operational metrics, revenue turnover, and an interactive 16-hour slot grid.

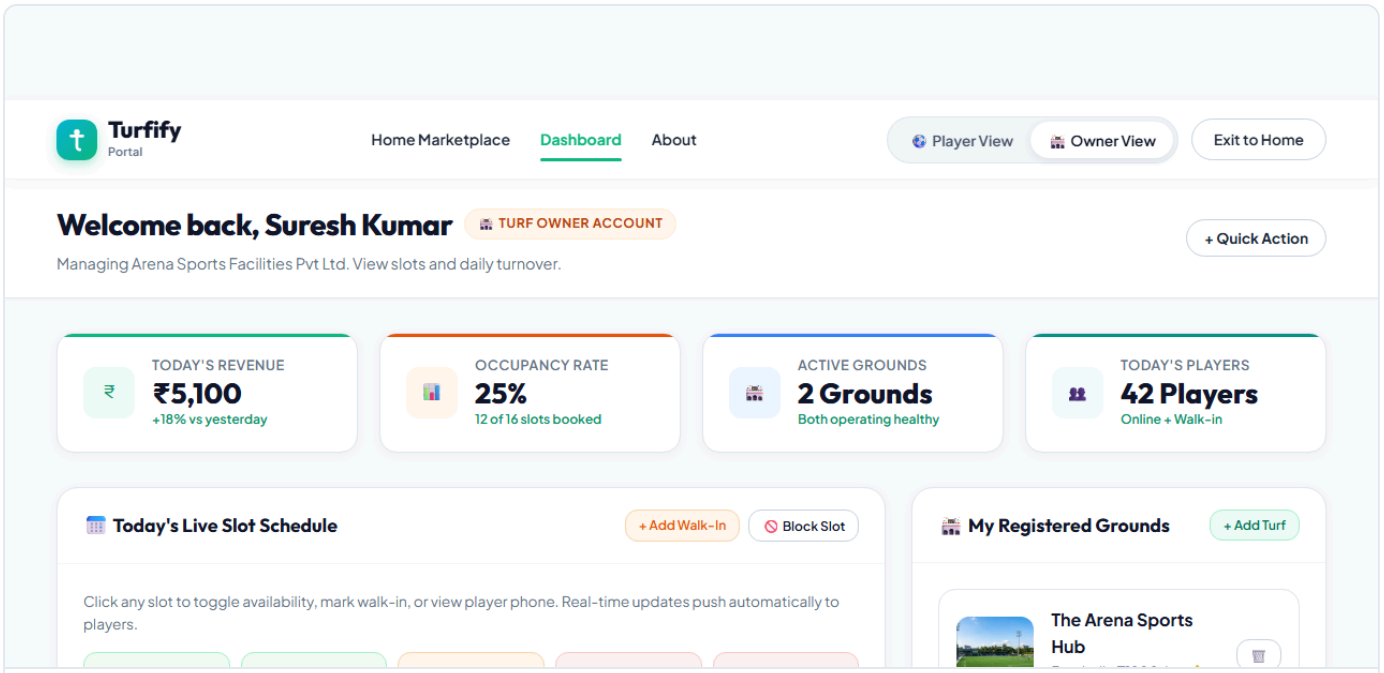


Figure 6.1 — Turf Owner Partner Console with Live Turnover, Occupancy Matrix, and Walk-in Blocker OWNER ERP

Today's Turnover +18% VS YESTERDAY

₹9,600

Gross revenue collected online + cash at ground.

Slot Occupancy 12 OF 16 BOOKED

75% Capacity

Booked hours vs total daily operational facility slots.

6.1 Interactive Slot Matrix & Walk-in Cash Management

Slot State	Visual Indicator	What Happens on Click
Available Slot	OPEN (GREEN)	Opens **Walk-in Booking Modal**. Owner logs cash customer; instantly blocks slot on the marketplace to prevent double booking.
Online Booking	PAID ONLINE (ORANGE)	Reveals player contact details (Rahul Sharma, +91 98765 43210, #BK-101) so manager can contact team captain.
Maintenance Slot	MAINTENANCE (DARK)	Used during turf brushing, lighting repair, or private coaching. Blocks public marketplace bookings.

7. Dual-Role Auth Engine & Master Interaction Matrix

Turfify supports unified single-identity authentication where a user can operate as a player or switch into turf facility ownership.

⚽ Player Authentication

CONSUMER FLOW

1. User clicks **"Login"** in navbar.
2. Selects radio button "⚽ **Player Account**".
3. Submits form → Authenticates via `TurfifyStore.switchRole('player')`.
4. Toast confirms login and redirects to `player-dashboard.html`.

🏟️ Turf Owner Partner Flow

ENTERPRISE FLOW

1. User clicks **"List Your Turf"** or **"Register"**.
2. Selects radio button "🏟️ **Turf Owner**".
3. Submits form → Authenticates via `TurfifyStore.switchRole('owner')`.
4. Toast confirms login and redirects to `owner-dashboard.html`.

7.1 Master Interaction Summary Reference

Page	Button / Trigger ID	User Input	Frontend Screen Output
`index.html`	`#select-location`	Picks area from dropdown	Filters turf listing cards; triggers match count toast.
`index.html`	`.sport-pill-card`	Clicks sport icon card	Filters sport venues; smooth scrolls down to venue grid.
`index.html`	`.btn-card-book`	Clicks "Book Now" on turf	Opens Booking Modal with pre-filled rates and time slots.
`index.html`	`.slot-btn`	Clicks an hourly slot	Highlights green; calculates total pricing.
`index.html`	`#btn-confirm-slot`	Clicks "Confirm & Pay"	Commits reservation to store; shows toast; updates hub.
`index.html`	`.btn-card-heart`	Clicks heart icon	Toggles active red heart; adds venue to favorites.
`index.html`	`#btn-hero-owner`	Clicks "Register Your Turf"	Opens facility onboarding modal for turf owners.
`player-dashboard`	`.btn-match-primary`	Clicks "Get Directions"	Launches Google Maps GPS navigation to pitch.
`player-dashboard`	`.btn-match-secondary`	Clicks "Share Squad Link"	Copies WhatsApp squad invite token to clipboard.
`owner-dashboard`	`#btn-open-add-turf`	Clicks "+ Add New Ground"	Opens facility registration modal to add additional pitches.
`dashboard.html`	`#nav-btn-owner`	Clicks "Owner View"	Instantly toggles dashboard view into partner console.

Document Status: Production Certified & Client Ready

All frontend interactions match the production codebase in `c:/Users/shame/abu`.

PRODUCTION CERTIFIED